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Innovation Education: Restarting Conversations about How We Teach, Educate, and Train Innovation Practitioners

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In the world of disruptions and generative artificial intelligence (AI), innovation, R&D, and technology management practitioners must navigate, design, and even lead complex innovation ecosystems and processes. Effectively managing these demands requires the deployment of technical and non-technical skills. While these arduous activities increase the necessity for firms to deepen research and technological collaboration with higher education institutions (Etzkowitz et al. 2000), there is also a need for industry to have more involvement in educational programs. Since these educational programs at higher education institutions train the next generation of innovation, R&D, and technology management practitioners, it makes sense that firms should pay more attention to these programs, right?

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Historically, a notable divide existed between the academic world and industry. Academics often felt industry didn't understand research, while industrial research practitioners sometimes viewed academics as having their heads in the clouds (Harris 1968). Despite this perceived separation, the relationship between industry and educational institutions has always been an important subject for policy makers, practitioners, and academics alike. Various initiatives were introduced to strengthen this relationship. For instance, in the mid 1960s, various formal relationships were identified, including university teacher–graduate student teams working at company laboratories and companies employing research advisory committees from universities (Cuthbert and Konig 1965). These interfaces were necessary, as a significant portion of technically trained students would end up in industry, working in small, medium, and large enterprises (Harris 1968). Similarly, researcher-in-residence programs have been used to host academics in companies and practitioners in higher education institutions as knowledge transfer conduits to understand changing needs (Graver and Stamos 1997; Jones and Craven 2001).

Innovation, R&D, and technology management practice often draw upon engineering, science, and management knowledge to plan, develop, and implement technological capabilities to shape and accomplish an organization's strategic and operational objectives (Weimer 1991). Hence, educating individuals for these roles requires specific requirements. For example, ensuring graduates have an understanding of integrating technology into strategic objectives, accelerating entry into and exit from technologies, assessing technology efficiently, and accomplishing technology transfer.

Developing these capabilities and skills in the next generation of practitioners is important. Educational institutions and firms play significant roles in shaping how these skills and capabilities are imparted to graduates (Heaton, Siegel, and Teece 2019). New innovation concepts and practices that alter how technologies are developed, managed, and commercialized—such as open innovation, distributed innovation, and value creation networks—further exacerbate the need for educational institutions to modify what they teach (Nambisan and Wilemon 2004).

One such capability is problem-solving under uncertainty, which is crucial as most companies operate in a VUCA (volatility, uncertainty, complexity, and ambiguity) world. In an interview (Miller and Euchner 2014), Richard Miller, the founding president of Olin College of Engineering, suggested that traditional teaching approaches, often focused on rote learning, analysis, and the application of complex mathematics to well-defined problems, may not fully engage students or prepare them for the creative act of "innovating" and solving real-world, often ill-defined, problems. There's a need to cultivate and hone multiple soft and nontechnical capabilities and skills beyond the purely technical. The future of engineering education should aim to train students in the 21st-century skills required by innovation practitioners. This involves moving beyond just knowing the "scales" (like calculus and physics) to being able to "improvise" through deep, dynamic collaboration.

For practitioners already in industry, a major challenge is the rapid pace of technological advancement and knowledge growth, leading to the potential obsolescence of prior knowledge. The ability to act under such conditions increasingly involves an appreciation of the systems in which technologies and firms are embedded. Innovation practitioners, therefore, need ongoing education to recognize, appreciate, and work with (and within) the entrepreneurial and innovation ecosystems that can support new venture formation and company growth (Galanakis 2006; Kapsali 2011). The question of "updating" technical personnel through continuing education and training has been a focus of human resources and organizational development academics and practitioners for decades. However, for those of us in the innovation, R&D, and technology management space, this focus has been sporadic but is gaining traction in recent work (Lythreatis, Singh, and El-Kassar 2022). The consensus here is that effective teaching and learning in innovation, R&D, and technology managements fields require robust cooperation between industry and academia. This partnership is essential for defining curriculum, delivering content, and supporting research and educational initiatives (Nambisan and Wilemon 2004).

Collaborations in this area are ongoing and thriving. Some anecdotal examples include engaging industry stakeholders when designing educational programs, inviting industry guest speakers to lecture in class, internships and industry placements, and students working on actual problems identified by organizations in project-based courses. What remains unanswered includes whether there are new ways to collaborate on educational programs; how academics are teaching new skills and capabilities, such as the adoption of generative AI; how organizations are training and upskilling their employees; and how educational institutions and organizations instill an entrepreneurial and innovative mindset and culture.

This is by no means an exhaustive list. But now is the time for practitioners and academics to reflect on educating, teaching, and learning within the innovation, R&D, and technology management fields. Addressing these critical questions demands more than continuing current efforts; it requires a proactive and imaginative partnership between academia and industry.

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